

Physical Identification of $SU(3)_c$ and the Fundamental Representation **3** from the PDL Axioms C1–C4

Projective Dynamic Logo Framework — Document D59

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Abstract

Document D58 [1] established $SU(3)$ as an unconditional algebraic theorem of the PDL axioms C1–C4, via the Weyl group $S_3 = S_4/V_4$, the centre $\mathbb{Z}_3 = A_4/V_4$, and the A_2 root system in the trace-zero plane. That derivation identified the group but left open the question of its physical realisation: on which three-dimensional space does $SU(3)$ act as the colour gauge group $SU(3)_c$ of quantum chromodynamics? The present document resolves this question by constructing the fundamental representation **3** of $SU(3)$ explicitly from C1–C4. The carrier space is identified as the trace-zero plane $W = \{(a, b, c) \in \mathbb{C}^3 : a + b + c = 0\}$, whose three natural axes are canonically identified with the three elements of $V_4 \setminus \{e\}$ — the structural triplet of K_4 established in D58 Lemma L2. The derivation proceeds through five lemmas verified by exhaustive exact-integer computation in the supplementary script `PDL_D59_script1.py` [2]: (L1) the S_3 -action on $V_4 \setminus \{e\}$ is faithful and transitive (recalled from D58); (L2) the character of the induced permutation representation on $\mathbb{C}^3$ decomposes as $\chi_\pi = \chi_{\text{trivial}} + \chi_{\text{standard}}$; (L3) the trace-zero subspace W is the unique S_3 -invariant complement of the trivial line $\mathbb{C}(1, 1, 1)$ in $\mathbb{C}^3$; (L4) the A_2 root system is realised in W with the six roots permuted transitively by S_3 and Cartan matrix $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$; (L5) the canonical ordering $k_1 < k_2 \ll k_3$ of the leakage-cycle indices (D51), combined with the isospin asymmetry $\Delta n = n_d - n_u = 4 > 0$ forced by C4 (D47), induces a canonical orientation on the three axes of W that selects the representation **3** over its conjugate $\bar{\mathbf{3}}$, identifying matter with the colour-charge sector. A documented negative result shows that the bijection between the leakage-cycle triplet (T_2) and the orbit triplet (T_3) is physically motivated by the quintuplet, but not S_4 -equivariant. Together with D46 [3], D57 [4], and D58 [1], D59 completes the derivation of the colour gauge group with its fundamental representation from first principles, without free parameters beyond Δm_{iso} .

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1 Introduction and context

1.1 What D58 achieved and what remained open

Document D58 [1] closed the algebraic derivation of the Standard Model gauge group within the PDL programme. Starting from four combinatorial axioms on finite signed graphs — with no presupposed spacetime, no continuum, and no field content — it established the group $SU(3)$ as an unconditional theorem of C1–C4, via the following chain of five lemmas:

1. $S_4/V_4 \cong S_3$ (Weyl group of A_2),
2. equivariant bijection $V_4 \setminus \{e\} \leftrightarrow$ three V_4 -orbits on edges of K_4 ,
3. $A_4/V_4 \cong \mathbb{Z}_3$ (centre of $SU(3)$),
4. rank-2 Cartan subalgebra forced by $R_e = 6$,
5. A_2 root system in the trace-zero plane.

Combined with D46 [3] ($U(1)$) and D57 [4] ($SU(2)$), D58 established the algebraic structure $SU(3) \times SU(2) \times U(1)$ as an unconditional theorem. However, three open problems were identified in D58:

- OP-D58-1: the explicit identification of $SU(3)$ with the physical colour gauge group $SU(3)_c$, acting on a three-dimensional space constructed from the PDL axioms;
- OP-D58-2: the precise relationship between the three triplets of K_4 — cycle-homology triplet (T1), leakage-cycle triplet (T2), orbit triplet (T3);
- OP-D58-3: the derivation of the fundamental representation **3** of $SU(3)$ from C1–C4.

The present document addresses OP-D58-1 and OP-D58-3 directly, and clarifies OP-D58-2 by a documented negative result.

1.2 The central question

Knowing that $SU(3)$ exists as a theorem is not the same as knowing how it acts. In the Standard Model, $SU(3)_c$ acts on quarks in the fundamental representation **3**: each quark carries a colour charge taking values in a three-dimensional complex space, and the eight gluons mediate transitions between colour states. None of this action is contained in the algebraic existence theorem of D58.

The central question of D59 is: *does the PDL formalism, via C1–C4, contain a canonical three-dimensional complex vector space on which $SU(3)$ acts in the representation **3**, and if so, which?*

The answer is yes. The carrier space is the *trace-zero plane* $W = \{(a, b, c) \in \mathbb{C}^3 : a+b+c = 0\}$, whose three natural coordinate axes are canonically identified with the three elements of $V_4 \setminus \{e\}$. The action of $S_3 = S_4/V_4$ on $V_4 \setminus \{e\}$ — already established in D58 as the Weyl group action on the three simple roots of A_2 — extends uniquely to an irreducible action of $SU(3)$ on W , and this is precisely the fundamental representation **3**.

1.3 Why this is not trivial

One might expect that, having derived $SU(3)$, its fundamental representation would follow automatically. This is not the case for the following reason. The Lie algebra $\mathfrak{su}(3)$ admits two inequivalent fundamental representations of dimension 3: the representation $\mathbf{3}$, acting on the carrier space of quarks, and its conjugate $\bar{\mathbf{3}}$, acting on antiquarks. These two representations are related by complex conjugation and are physically distinct: $\mathbf{3}$ couples to matter and $\bar{\mathbf{3}}$ to antimatter. A derivation from first principles must therefore provide a structural reason for preferring one orientation over the other.

In the PDL framework, this reason exists and is a theorem. The isospin asymmetry $\Delta n = n_d - n_u = 4 > 0$ (established as an unconditional theorem of C4 in D47 [5]) forces a canonical orientation on the three-element set $V_4 \setminus \{e\}$, selecting $\mathbf{3}$ over $\bar{\mathbf{3}}$ and thereby identifying the matter sector without an additional postulate.

1.4 Position in the broader literature

The derivation of gauge group *representations* from first principles is substantially harder than the derivation of the gauge *group* itself. Existing programmes — grand unification, string theory, noncommutative geometry, loop quantum gravity — either derive the group structure (sometimes) or read the representations from phenomenological input. No programme known to the author derives a specific gauge representation from a purely combinatorial or logical foundation prior to the present work.

The PDL result should be understood in proportion: it establishes the representation $\mathbf{3}$ on a three-dimensional space constructed from the axioms, and identifies the matter orientation from an integer forced by C4. It does not yet derive the full fermionic content of the Standard Model (quark masses, generation structure, CKM mixing). Those questions remain open and are identified explicitly in Section 12.

2 Prerequisites and notation

2.1 PDL quantities recalled

The following quantities from earlier corpus documents are used throughout. All are unconditional theorems of C1–C4; proofs are in the cited documents.

Definition 2.1 (Proton quintuplet [5, 6]). The PDL proton is characterised by the integer quintuplet

$$(n_u, n_d, r_{\text{val}}, R_{\text{sea}}, R_{\text{tot}}) = (24, 28, 930, 10087, 11017),$$

where $n_u = 24$ (u -core), $n_d = 28$ (d -core), $r_{\text{val}} = 930$ (valence budget), $R_{\text{sea}} = 10087$ (sea budget), $R_{\text{tot}} = 11017$ (total budget). The PDL electron prototype K_4 has relational budget $R_e = |E(K_4)| = 6$.

Definition 2.2 (Isospin asymmetry [5]). The isospin asymmetry $\Delta n = n_d - n_u = 4$ is an unconditional theorem of C4. It is the unique value for which the discriminant of the quasi-completeness equation is a perfect square, yielding a positive integer solution $(n_u, n_d) = (24, 28)$.

Definition 2.3 (Leakage-cycle indices [7]). The three coupling-mode indices

$$\begin{aligned} k_1 &= n_d - n_u + R_e - 1 = 9, \\ k_2 &= n_u - R_e + 1 = 19, \\ k_3 &= R_e \cdot n_d = 168, \end{aligned}$$

are integer theorems of C1–C4 via the quintuplet. They satisfy the completeness identity $k_1 + k_2 = n_d = 28$ and the ratio identity $k_3/(k_1 + k_2) = R_e = 6$, both exact. The primes at positions k_1, k_2, k_3 in the prime sequence are $(p_{k_1}, p_{k_2}, p_{k_3}) = (23, 67, 997)$, which determine the PDL cosmological constant (D51–D52 [7, 8]).

Definition 2.4 (The three triplets of K_4 [7, 1]). Three distinct triplets are associated with K_4 :

- (T_1) : three independent cycles in $H_1(K_4; \mathbb{Z})$, spanning the first homology of K_4 ;
- (T_2) : the leakage-cycle triplet $\{k_1, k_2, k_3\} = \{9, 19, 168\}$, or equivalently the prime triplet $(23, 67, 997)$;
- (T_3) : the three elements of $V_4 \setminus \{e\}$, equivalently the three V_4 -orbits on the six edges of K_4 , equivalently the three partitions of $\{0, 1, 2, 3\}$ into two unordered pairs.

2.2 Results from D58 recalled

Proposition 2.5 (D58, Lemmas L1–L2 [1]). *The quotient S_4/V_4 is isomorphic to S_3 . The conjugation action of S_4 on $V_4 \setminus \{e\}$ descends to a faithful transitive action of $S_3 = S_4/V_4$ on $V_4 \setminus \{e\}$, with kernel V_4 . The three elements of $V_4 \setminus \{e\}$ are in canonical S_4 -equivariant bijection with the three V_4 -orbits on the six edges of K_4 .*

Proposition 2.6 (D58, Lemma L3 [1]). *The quotient $A_4/V_4 \cong \mathbb{Z}_3$ is the centre of $\text{SU}(3)$. Under the homomorphism $\rho: S_4 \rightarrow S_3$, the image of A_4 is the alternating subgroup $\text{Alt}(3) \cong \mathbb{Z}_3 \subset S_3$.*

Proposition 2.7 (D58, Lemmas L4–L5 and main theorem [1]). *The unique compact simply connected simple Lie group with Weyl group S_3 , centre $\mathbb{Z}_3$, rank-2 Cartan subalgebra (forced by $R_e = 6$), and A_2 root system is $\text{SU}(3)$. This group is an unconditional theorem of C1–C4 modulo the classical Cartan–Killing–Weyl classification.*

3 Lemma L1: the S_3 -action on $V_4 \setminus \{e\}$

Lemma: L1 — Faithful transitive action (recalled from D58)

The group $S_3 = S_4/V_4$ acts faithfully and transitively on the three-element set $V_4 \setminus \{e\}$ by conjugation: for $[g] \in S_3$ and $v \in V_4 \setminus \{e\}$, the action is $[g] \cdot v = gvg^{-1}$. The kernel of this action is trivial, and all three elements of $V_4 \setminus \{e\}$ lie in a single orbit.

Proof. This is Lemma L2(a)(b)(c) of D58 [1], verified exhaustively over all 24 elements of S_4 in `PDL_SU3_script1.py` [9]. It is re-verified in the present document’s script (`PDL_D59_script1.py` [2], checks L1a–L1c) for self-containedness. The three elements are $v_0 = (01)(23)$, $v_1 = (02)(13)$,

$v_2 = (03)(12)$; the six non-trivial conjugations by elements of S_4 produce all six permutations of $\{v_0, v_1, v_2\}$. $\square$

Remark 3.1. Lemma L1 identifies S_3 as a concrete permutation group on the set $\{v_0, v_1, v_2\}$. This is the starting point for the construction of the carrier space W .

4 Lemma L2: the permutation representation and its character

Lemma: L2 — Character decomposition of the permutation representation

Let $\pi: S_3 \rightarrow \text{GL}(\mathbb{C}^3)$ be the permutation representation induced by the S_3 -action on $\{v_0, v_1, v_2\}$ via Lemma L1. The character of π takes the values

$$\chi_\pi(e) = 3, \quad \chi_\pi(\text{transposition}) = 1, \quad \chi_\pi(3\text{-cycle}) = 0,$$

and decomposes as

$$\pi \cong \mathbf{1} \oplus \mathbf{2},$$

where $\mathbf{1}$ is the trivial representation of S_3 and $\mathbf{2}$ is the unique two-dimensional irreducible representation (the standard representation).

Proof. The character of the permutation representation is the number of fixed points of each group element. In S_3 : the identity fixes all three elements ($\chi = 3$); each transposition fixes exactly one element ($\chi = 1$, confirmed for all 12 odd elements of S_4 in the script); each 3-cycle fixes no element ($\chi = 0$, confirmed for all 8 elements of $A_4 \setminus V_4$). The decomposition into irreducibles follows from the character table of S_3 : the trivial character $\chi_1 = (1, 1, 1)$ and the standard character $\chi_2 = (2, 0, -1)$ (for conjugacy classes e , transpositions, 3-cycles) satisfy $\chi_\pi = \chi_1 + \chi_2$, as $(3, 1, 0) = (1, 1, 1) + (2, 0, -1)$. Verified exhaustively over all 24 elements of S_4 in `PDL_D59_script1.py` (checks L2a–L2e). $\square$

5 Lemma L3: the trace-zero plane as the carrier space

Lemma: L3 — The trace-zero plane W is the carrier of $\mathbf{2}$

In the permutation representation π on $\mathbb{C}^3$ with basis $\{e_{v_0}, e_{v_1}, e_{v_2}\}$, the vector $(1, 1, 1)$ spans the unique S_3 -invariant line (carrier of $\mathbf{1}$), and its orthogonal complement

$$W = \{(a, b, c) \in \mathbb{C}^3 : a + b + c = 0\}$$

is the unique S_3 -invariant two-dimensional subspace (carrier of $\mathbf{2}$). The six matrices of S_3 restricted to W form a faithful representation by 2×2 integer matrices of determinant ± 1 , all of which are distinct.

Proof. The vector $(1, 1, 1)$ is fixed by all permutation matrices (every element of S_3 permutes the coordinates, preserving their sum), so it spans the trivial subspace. The subspace $W = \ker(\sum_i)$

is preserved by all permutations since $a + b + c = 0$ implies $a_{\sigma(0)} + a_{\sigma(1)} + a_{\sigma(2)} = 0$ for any permutation σ . Since W is two-dimensional and $\mathbb{C}^3 = \mathbb{C}(1, 1, 1) \oplus W$ as S_3 -modules (by Maschke's theorem, since $|S_3| = 6$ is invertible over $\mathbb{C}$), W carries the complement **2**. The six restricted matrices, expressed in the basis $\{e_1 = (1, -1, 0), e_2 = (0, 1, -1)\}$ of W , are verified to be six distinct integer matrices with $|\det| = 1$ in `PDL_D59_script1.py` (checks L3a–L3e). $\square$

Remark 5.1 (The carrier space is forced by K_4). The space W is not postulated. Its three natural axes $\{e_{v_0}, e_{v_1}, e_{v_2}\}$ are labelled by the three elements of $V_4 \setminus \{e\}$, which are themselves the three V_4 -orbits on the edges of K_4 (D58, Lemma L2). The trace-zero constraint $a + b + c = 0$ is the unique S_3 -invariant condition on $\mathbb{C}^3$, forced by the group action derived from C1–C4. The carrier space is therefore a theorem of the axioms.

6 Lemma L4: the A_2 root system in W

Lemma: L4 — A_2 root system in the trace-zero plane

The six vectors

$$\Phi = \{(1, -1, 0), (-1, 1, 0), (0, 1, -1), (0, -1, 1), (1, 0, -1), (-1, 0, 1)\} \subset W$$

form the A_2 root system in W . They satisfy:

- (i) all six have squared Euclidean norm $\|\alpha\|^2 = 2$;
- (ii) the S_3 -action on $V_4 \setminus \{e\}$ permutes Φ transitively (single orbit);
- (iii) Φ is closed under negation: $\alpha \in \Phi \Rightarrow -\alpha \in \Phi$;
- (iv) all pairwise inner products lie in $\{-2, -1, 0, 1, 2\}$;
- (v) the Cartan matrix of the simple roots $\alpha_1 = (1, -1, 0)$ and $\alpha_2 = (0, 1, -1)$ is $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$, the Cartan matrix of A_2 .

Proof. Properties (i)–(v) are verified by direct integer computation in `PDL_D59_script1.py` (checks L4a–L4f). The transitivity of the S_3 -action (ii) follows from the fact that S_3 acts on W by 2×2 integer matrices of determinant ± 1 (Lemma L3), and a direct orbit computation confirms that starting from any root and applying all six elements of S_3 recovers all six roots. $\square$

Remark 6.1. Lemma L4 confirms that the root system of $SU(3)$ is realised concretely in the plane W whose axes are labelled by $V_4 \setminus \{e\}$. The three pairs $\{\pm\alpha_i\}$ of roots correspond, via the S_3 -orbit structure, to the three elements of $V_4 \setminus \{e\}$. The A_2 Cartan matrix — whose unique compact simply connected simple Lie group is $SU(3)$ by the Cartan–Killing–Weyl classification — is thus a direct arithmetic consequence of the edge structure of K_4 (via $R_e = 6$, which determined the rank-2 Cartan in D58 Lemma L4).

6.1 Connection to the Weyl group

The standard representation $\mathbf{2}$ of S_3 on W is precisely the reflection representation of the Weyl group $W(A_2) \cong S_3$: the three transpositions in S_3 act on W as reflections in the three root hyperplanes $\langle \alpha_i \rangle^\perp$. This is not a coincidence imposed from outside: it is the direct consequence of the fact that the S_3 -action on $V_4 \setminus \{e\}$ was derived from the automorphism group S_4 of K_4 via C1–C4, and the root hyperplane structure emerges from the same combinatorial data.

7 Lemma L5: orientation, $\mathbf{3}$ versus $\bar{\mathbf{3}}$, and the isospin asymmetry

7.1 The two inequivalent fundamental representations

The Lie algebra $\mathfrak{su}(3)$ admits two inequivalent fundamental representations of dimension 3. They are related by complex conjugation:

- $\mathbf{3}$: the defining representation, acting on column vectors of $\mathbb{C}^3$ by matrix multiplication $U \mapsto U\mathbf{v}$;
- $\bar{\mathbf{3}}$: its conjugate, acting by $U \mapsto \bar{U}\mathbf{v}$, equivalently by $(U^{-1})^T$.

In quantum chromodynamics, $\mathbf{3}$ acts on quarks and $\bar{\mathbf{3}}$ on antiquarks. The two representations are distinguished by the orientation of the $\mathbb{Z}_3$ centre action: on $\mathbf{3}$, the centre element $e^{2\pi i/3}\mathbf{1}$ acts as $e^{2\pi i/3}$; on $\bar{\mathbf{3}}$, it acts as $e^{-2\pi i/3}$. At the level of the Weyl group $S_3 \cong W(A_2) \subset \mathrm{SU}(3)$, the two representations differ by the choice of the cyclic orientation on the set $\{v_0, v_1, v_2\}$: the orientation $(v_0 \rightarrow v_1 \rightarrow v_2 \rightarrow v_0)$ corresponds to $\mathbf{3}$, and the opposite orientation $(v_0 \rightarrow v_2 \rightarrow v_1 \rightarrow v_0)$ corresponds to $\bar{\mathbf{3}}$.

Lemma: L5 — Canonical orientation from the isospin asymmetry

The canonical ordering $k_1 < k_2 < k_3$, forced by the quintuplet via Definition 2.3, labels the three elements of $V_4 \setminus \{e\}$ as (v_0, v_1, v_2) in increasing order. The isospin asymmetry $\Delta n = n_d - n_u = 4 > 0$ (Theorem of C4, Definition 2.2) forces $k_1 < k_2$, i.e. the ordering is strict and positive. The cyclic orientation $(v_0 \rightarrow v_1 \rightarrow v_2 \rightarrow v_0)$ induced by this ordering is preserved by the $\mathbb{Z}_3 = A_4/V_4$ centre action (even permutations of S_3) and reversed by the transpositions (odd permutations). This selects the representation $\mathbf{3}$ over $\bar{\mathbf{3}}$.

Proof. The ordering $k_1 < k_2 < k_3$ (i.e. $9 < 19 < 168$) is a theorem of C1–C4 via the quintuplet (check L5b of the script). The positivity $\Delta n > 0$ is the content of D47, Theorem OP13 (check L5a). The image of A_4 in $\mathrm{Sym}(V_4 \setminus \{e\})$ is exactly $\mathrm{Alt}(3) = \{(v_0 v_1 v_2), (v_0 v_2 v_1), \mathrm{id}\}$ (check L5c, recalled from D58 Lemma L3). The generator $(v_0 \rightarrow v_1 \rightarrow v_2)$ of $\mathrm{Alt}(3)$ maps the cyclic order (v_0, v_1, v_2) to (v_1, v_2, v_0) , which is the same cycle (check L5d–L5e). Hence the cyclic orientation $(v_0 \rightarrow v_1 \rightarrow v_2 \rightarrow v_0)$ is invariant under the $\mathbb{Z}_3$ centre action, as required for the representation $\mathbf{3}$.

The ratio identity $k_3/(k_1 + k_2) = R_e \cdot n_d/n_d = R_e = 6$ (exact, check L5f) confirms that k_3 carries the full d -coupling surface of K_4 , while $k_1 + k_2 = n_d$ accounts for the complete d -sector. The asymmetry $\Delta n = n_d - n_u = 4 > 0$ means the d -sector strictly exceeds the u -sector,

inducing the strict ordering $k_1 < k_2$ (since $k_1 = \Delta n + R_e - 1$ and $k_2 = n_u - R_e + 1$, and $k_1 < k_2 \Leftrightarrow \Delta n < n_u - 2R_e + 2 = 14$, which holds). The positivity $\Delta n > 0$ (check L5g) is the structural origin of the orientation, and identifies the representation $\mathbf{3}$ with the matter sector. $\square$

Remark 7.1 (Physical interpretation). In quantum chromodynamics, the distinction between quarks ($\mathbf{3}$) and antiquarks ($\bar{\mathbf{3}}$) is imposed by hand via the charge-conjugation parity of the fields. In the PDL framework, this distinction emerges from the integer $\Delta n = 4$, which is itself the unique value compatible with the quasi-completeness equation of the proton (D47). The matter/antimatter asymmetry of the colour sector has, within PDL, a combinatorial origin: the proton cannot exist with $\Delta n \leq 0$, and its existence forces the orientation of the colour-charge space.

8 The main theorem

Under the PDL axioms C1–C4 and the classical Cartan–Killing–Weyl classification of compact simple Lie groups:

- (i) The three-dimensional complex vector space $W = \{(a, b, c) \in \mathbb{C}^3 : a + b + c = 0\}$, with axes canonically labelled by $V_4 \setminus \{e\}$ and the $S_3 = S_4/V_4$ action from D58, is an unconditional consequence of C1–C4.
- (ii) The restriction of the S_3 -action on $V_4 \setminus \{e\}$ to W realises the standard two-dimensional irreducible representation of the Weyl group $W(A_2) \cong S_3$ of $\text{SU}(3)$, and extends uniquely (up to equivalence) to the fundamental representation $\mathbf{3}$ of $\text{SU}(3)$ on $W \cong \mathbb{C}^2$.
- (iii) The isospin asymmetry $\Delta n = 4 > 0$ (unconditional theorem of C4, D47) selects the orientation $\mathbf{3}$ over $\bar{\mathbf{3}}$, identifying the matter sector from the axioms without additional input.
- (iv) The physical colour gauge group $\text{SU}(3)_c$, acting on the colour-charge space W in the fundamental representation $\mathbf{3}$, is an unconditional theorem of C1–C4 modulo the Cartan–Killing–Weyl classification.

Proof. Part (i): by Lemmas L1–L3, the space W is determined by the S_3 -action on $V_4 \setminus \{e\}$ (itself an unconditional theorem of D58/C1–C4), and by the unique invariant decomposition $\mathbb{C}^3 = \mathbb{C}(1, 1, 1) \oplus W$. Part (ii): by Lemma L4, the A_2 root system is realised in W with the correct Cartan matrix. By the Cartan–Killing–Weyl classification, the unique compact simply connected simple Lie group with Weyl group S_3 and A_2 root system is $\text{SU}(3)$ (established in D58). The standard representation of $W(A_2) \cong S_3$ on W extends to the representation $\mathbf{3}$ of $\text{SU}(3)$, because $\mathbf{3}$ is the unique irreducible 2-dimensional (as a real $\mathfrak{su}(3)$ -module it is 3-complex-dimensional, but its restriction to the Weyl group is exactly the standard representation, as follows from the weight space decomposition of $\mathbf{3}$ under the Cartan of $\mathfrak{su}(3)$). Part (iii): by

Lemma L5, the canonical orientation is forced by $\Delta n > 0$ (D47, unconditional). Part (iv): follows from (i)–(iii) by definition of $SU(3)_c$ as $SU(3)$ acting in the fundamental representation $\mathbf{3}$ on the colour-charge space. $\square$

8.1 The complete causal chain

$$\begin{aligned}
\text{C1–C4} &\longrightarrow K_4 \longrightarrow S_4 = \text{Aut}(K_4) \\
&\longrightarrow S_4/V_4 \cong S_3 = W(A_2) \quad [\text{D58, L1}] \\
&\longrightarrow SU(3) \quad [\text{D58, Cartan–Killing–Weyl}] \\
&\longrightarrow W = \{a + b + c = 0\} \subset \mathbb{C}^3 \quad [\text{D59, L2–L3}] \\
&\longrightarrow A_2 \text{ root system in } W \quad [\text{D59, L4}] \\
&\longrightarrow \Delta n = 4 > 0 \text{ selects } \mathbf{3} \text{ over } \bar{\mathbf{3}} \quad [\text{D59, L5 from D47}] \\
&\longrightarrow SU(3)_c \text{ acting on } W \text{ in } \mathbf{3}. \quad [\text{D59, Theorem}]
\end{aligned}$$

Every arrow is a theorem: arrows 1–3 are unconditional theorems of C1–C4 (established in D46–D58); arrow 4 invokes the classical Cartan–Killing–Weyl classification; arrows 5–7 are unconditional theorems of C1–C4 established in the present document.

9 The negative result: triplets (T2) and (T3) are not S_4 -equivariantly bijectable

The leakage-cycle triplet $(T_2) = \{k_1, k_2, k_3\} = \{9, 19, 168\}$ and the orbit triplet $(T_3) = V_4 \setminus \{e\}$ cannot be related by a canonical S_4 -equivariant bijection. Any bijection between them is physically motivated by the quintuplet structure, but not forced by group-theoretic equivariance.

Proof. The three elements of (T_3) carry a transitive S_4 -action (Lemma L1): any of the 6 elements of S_3 maps any element of $V_4 \setminus \{e\}$ to any other. In particular, the stabiliser of any single element is trivial in S_3 . By contrast, the three integers $\{k_1, k_2, k_3\} = \{9, 19, 168\}$ are three distinct values with no non-trivial symmetry under permutation: no non-identity element of S_3 fixes the ordered triple (k_1, k_2, k_3) as a set (verified in the script, check NEG-b). Hence any bijection $(T_2) \rightarrow (T_3)$ must break the S_3 -symmetry of (T_3) , and no such bijection is S_4 -equivariant. $\square$

Remark 9.1. This negative result is scientifically informative. It clarifies that $(T_3) = V_4 \setminus \{e\}$ is the *canonical* carrier of the S_3 -action and the A_2 root system, while (T_2) labels the same three objects from the perspective of the leakage dynamics (D51–D52). The canonical bijection between (T_2) and (T_3) is the index-ordering map $k_1 \leftrightarrow v_0, k_2 \leftrightarrow v_1, k_3 \leftrightarrow v_2$, which is motivated by the quintuplet structure (each index encodes a specific coupling-mode count involving n_u ,

n_d , and R_e) but is not forced by S_4 -equivariance. The relationship between (T_1) , (T_2) , and (T_3) remains OP-D58-2 in the corpus.

10 Computational verification

The complete derivation of D59 is accompanied by the verification script `PDL_D59_script1.py` [2], deposited separately on Zenodo. It provides exhaustive computational verification of every lemma using exact integer arithmetic and the Python standard library only (`fractions`, `itertools`). No floating-point is introduced.

Table 1: Items verified by `PDL_D59_script1.py`.

Verification	Method	Status
Quintuplet and leakage indices	Direct integer arithmetic	PASSED
L1a: S_4 stabilises $V_4 \setminus \{e\}$	Exhaustive conjugation, 24 elements	PASSED
L1b: image has 6 elements	Enumeration	PASSED
L1c: kernel = V_4	Identification, 4 elements	PASSED
L2a–e: character decomposition	Fixed-point count, 24 elements	PASSED
L3a: $(1, 1, 1)$ fixed by S_3	All 6 actions	PASSED
L3b–c: W preserved by S_3	Basis vectors, 6 actions	PASSED
L3d: 6 distinct 2×2 matrices on W	Explicit computation	PASSED
L3e: $ \det = 1$ for all matrices	Integer determinant	PASSED
L4a: norms ² = 2	Integer dot products	PASSED
L4b: S_3 permutes the 6 roots	Exhaustive	PASSED
L4c: transitive orbit	Orbit computation	PASSED
L4d: root pairing	Set membership	PASSED
L4e–f: Cartan matrix = A_2	Inner products	PASSED
L5a–g: orientation from $\Delta n > 0$	Integer inequalities	PASSED
NEG-a: K_i distinct	Trivial	PASSED
NEG-b: no S_3 symmetry on (T_2)	Exhaustive, 5 non-trivial perms	PASSED
All verifications		PASSED

In accordance with the PDL verrouillage protocol, no claim of D59 was incorporated into this document before independent execution of the verification script in Google Colab. The five lemmas and the negative result were verrouillés before LaTeX drafting began.

11 Epistemic status

Table 2: Complete epistemic stratification of D59 claims.

Claim	Status	Source
$\Delta n = 4$ unique (D47)	Theorem (C1–C4)	D47
$k_1 = 9, k_2 = 19, k_3 = 168$	Theorem (C1–C4)	D51
$k_1 + k_2 = n_d$	Theorem (exact)	D51
$k_3/(k_1 + k_2) = R_e = 6$	Theorem (exact)	D59, L5f
S_3 acts faithfully on $V_4 \setminus \{e\}$ (L1)	Theorem (C1–C4)	D58 L2, D59
$\chi_\pi = \chi_1 + \chi_2$ (L2)	Theorem (group theory)	D59, L2
$W = \{a + b + c = 0\}$ is S_3 -invariant (L3)	Theorem (C1–C4)	D59, L3
A_2 root system in W (L4)	Theorem (C1–C4)	D59, L4
Cartan matrix = $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$	Theorem (classical)	D59, L4
$\Delta n > 0$ selects 3 over $\bar{\mathbf{3}}$ (L5)	Theorem (C1–C4)	D59, L5 from D47
$SU(3)_c$ acts in 3 on W (main)	Theorem (classical)	D59, Theorem
$(T_2) \not\leftrightarrow (T_3)$ equivariantly	Theorem (verified)	D59, NEG
$(T_1), (T_2), (T_3)$ canonical relations	Open problem	OP-D58-2
Fermion mass spectrum from C1–C4	Open problem	OP-D59-1
Three colour generations from C1–C4	Open problem	OP-OFN-1

The label “Theorem (classical)” indicates that the proof additionally invokes the classical Cartan–Killing–Weyl classification and the representation theory of compact Lie groups, taken as established mathematical apparatus.

12 Open problems

Resolved by D59. The fundamental representation $\mathbf{3}$ of $SU(3)$ is constructed on the trace-zero plane $W \subset \mathbb{C}^3$, whose axes are labelled by $V_4 \setminus \{e\}$. The orientation selecting $\mathbf{3}$ over $\bar{\mathbf{3}}$ is forced by $\Delta n = 4 > 0$ (D47).

Resolved by D59. The derivation of $\mathbf{3}$ from C1–C4 is the content of Theorem 8.

Establish formally the relationship between (T_1) , (T_2) , and (T_3) (Definition 2.4). The negative result of Section 9 shows that no S_4 -equivariant bijection $(T_2) \leftrightarrow (T_3)$ exists; the precise nature of their relationship (via the quintuplet) remains to be formalised.
Priority: medium.

The present document establishes the carrier space W and the orientation $\mathbf{3}$ for the colour gauge group. It does not provide the quark masses m_u , m_d , m_s , etc. The single irreducible external parameter $\Delta m_{\text{iso}} = m_d - m_u$ enters PDL at the QCD interface (D31); the derivation of individual quark masses from C1–C4 is a distinct and harder problem.
Priority: medium.

The identification of the three leakage cycles with prime exponents (23, 67, 997) with the three generations of fermions, suggested by the common combinatorial origin $A_4/V_4 \cong \mathbb{Z}_3$ (D57–D58), remains open. The present document provides additional evidence for this identification (the same triplet (T_2) labels the canonical bijection with the axes of W), but a formal derivation requires a mechanism for the mass hierarchy between generations.
Priority: high. **Entry point:** N01, D57, D58, D59.

13 Conclusion

13.1 What D59 achieves

Document D59 completes the programme initiated in D46 and continued in D57–D58 by providing the first explicit construction, from the PDL axioms C1–C4, of a three-dimensional carrier space on which $SU(3)$ acts as the colour gauge group in the fundamental representation $\mathbf{3}$. The main results are:

1. The trace-zero plane $W = \{a+b+c = 0\} \subset \mathbb{C}^3$ is the canonical carrier of the **3** representation, with axes labelled by $V_4 \setminus \{e\}$ as forced by C1–C4 via K_4 .
2. The A_2 root system is realised in W with the Cartan matrix $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$, confirming at the geometric level what D58 established algebraically.
3. The isospin asymmetry $\Delta n = 4 > 0$ (unconditional theorem of C4, D47) selects the matter representation **3** over $\bar{\mathbf{3}}$ without additional input.
4. The ratio $k_3/(k_1 + k_2) = R_e = 6$ (exact integer) ties the leakage-cycle structure to the relational budget of K_4 , establishing an internal consistency between the cosmological (D51–D52) and gauge (D58–D59) sectors of the corpus.

13.2 Position within the PDL programme

The chain D46 + D57 + D58 + D59 now provides:

- U(1): U(1) gauge structure and Hopf fibration of K_4 (D46);
- SU(2): SU(2) gauge structure, Weinberg angle $\sin^2 \theta_W = 1/4$ at tree level (D57);
- SU(3): SU(3) gauge structure, algebraic derivation (D58);
- SU(3)_c in **3**: the colour gauge group with its fundamental representation, matter orientation identified (D59).

This constitutes the most complete derivation, from purely combinatorial axioms, of both the gauge group structure and a fundamental gauge representation currently available.

13.3 What remains open

D59 does not derive the Standard Model. The phenomenological content — quark masses, mixing angles, three fermion generations as dynamical replicas — remains open. The carrier space W is established; the full fermionic content requires additional structure. The programme advances slowly but with certainty.

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